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| <b>Name:</b> |  | <b>C.G:</b> |  |
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# ST ANDREW'S JUNIOR COLLEGE



**JC2 Preliminary Examination**

**H1 Chemistry (8872)**

**19<sup>th</sup> September 2011**

**Paper 2**

**2 hours**

Candidates answer Section A on the Question Paper

Additional materials: Data Booklet

Writing Paper

## READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

### Section A

Answer **all** the questions.

### Section B

Answer **two** questions on the writing paper provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
|--------------------|--|
| <b>Section A</b>   |  |
| <b>B6</b>          |  |
| <b>B7</b>          |  |
| <b>B8</b>          |  |
| <b>Total</b>       |  |

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This question paper consists of **17** printed pages including this page.

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### Section A (40 marks)

Answer **all** the questions in this section in the spaces provided.

- 1 The properties of beryllium, the first member of the alkaline earth metals, differ from the rest of the members of its group. Sharing a diagonal relationship with aluminium, beryllium usually behaves chemically similar to it.

- (a) (i) Beryllium chloride,  $\text{BeCl}_2$ , boils at  $520^\circ\text{C}$  while beryllium fluoride,  $\text{BeF}_2$ , boils at  $1169^\circ\text{C}$ .

Identify the bonding in  $\text{BeCl}_2$  and  $\text{BeF}_2$  respectively.

Account for the types of bonds present in both compounds.

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- (ii) In the solid state,  $\text{BeCl}_2$  molecules polymerise to make long chains.  
With the aid of a suitable labelled diagram, show the formation of its simple dimer which begins the polymerisation process.

- 1 (a) (iii) When solid beryllium chloride is heated above 215 °C, a vapour of  $M_r = 160.0$  is formed. When this vapour is heated above 1020 °C, another vapour of  $M_r = 80.0$  is formed instead.

At 645 °C, the average  $M_r$  of the vapour is found to be 126.0. Determine the molar composition of this vapour at 645 °C.

[5]

- (b) Both beryllium and aluminium metals could be extracted from the mineral, *Beryl*,  $(\text{Be}_3\text{Al}_2(\text{SiO}_3)_6)$ .

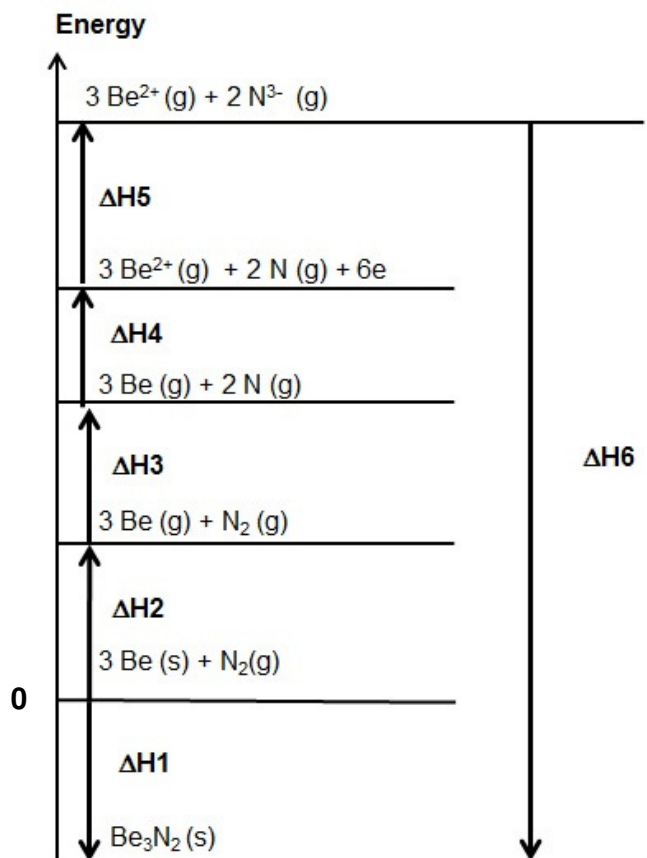
(i) Draw the dot-and-cross diagram of silicate,  $\text{SiO}_3^{2-}$ .

- (ii) When 5.33 tonnes of ore was mined, *Beryl* was found to be the only beryllium-containing mineral. Given that 30.6 kg of beryllium was extracted, determine the percentage of *Beryl* in the ore. (1 tonne = 1000 kg)

[3]

[Turn Over]

- 1 (c) Beryllium burns in nitrogen to form a refractory material, beryllium nitride,  $\text{Be}_3\text{N}_2$ . The following Born-Haber cycle shows the various energy conversions which occur for the formation of beryllium nitride.



- (i) Identify the following energy conversions:

|      |  |      |  |
|------|--|------|--|
| ΔH1: |  | ΔH3: |  |
| ΔH4: |  | ΔH6: |  |

- (ii) Using relevant information from the *Data Booklet* and the following information, calculate the lattice energy of beryllium nitride.

|                                                                                    |                                        |
|------------------------------------------------------------------------------------|----------------------------------------|
| $3\text{Be(s)} + \text{N}_2\text{(g)} \rightarrow \text{Be}_3\text{N}_2\text{(s)}$ | $\Delta H1 = -588 \text{ kJmol}^{-1}$  |
| $3\text{Be(s)} \rightarrow 3\text{Be(g)}$                                          | $\Delta H2 = +324 \text{ kJmol}^{-1}$  |
| $2\text{N(g)} + 6\text{e} \rightarrow 2\text{N}^{3-}\text{(g)}$                    | $\Delta H5 = +1404 \text{ kJmol}^{-1}$ |

- 1 (c) (iii) Explain if the lattice energy of lithium oxide is more or less exothermic than beryllium nitride.

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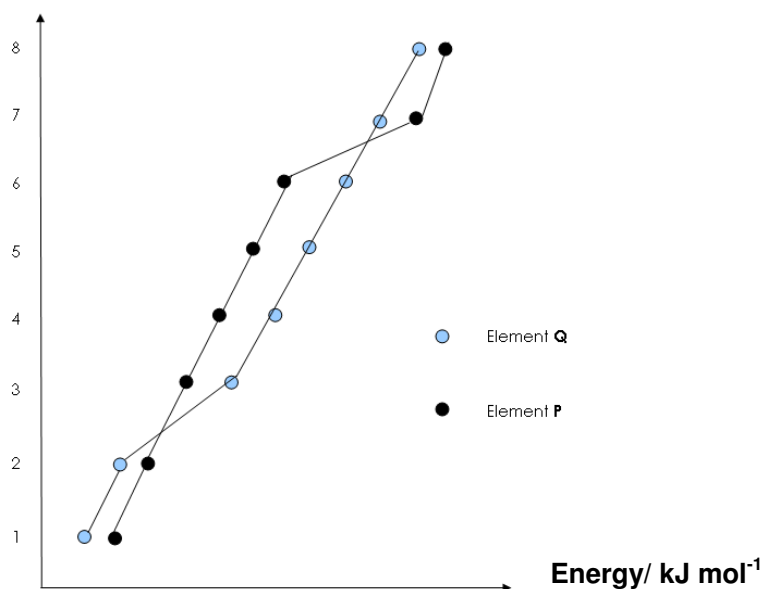
[7]

[Total marks: 15]

[Turn Over]

- 2 The diagram below shows the ionisation energies required to remove successive electrons from elements **P** and **Q**.

No. of electrons removed



- (a) Define first ionisation energy.

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[1]

- (b) (i) Identify the groups which elements **P** and **Q** belong to.

**P:** ..... **Q:** .....

- (ii) State two physical properties exhibited by the compound formed between **P** and **Q**.

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[3]

[Total marks: 4]

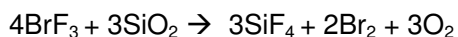
- 3 (a) Silicon(IV) oxide,  $\text{SiO}_2$ , is insoluble in water but it dissolves when  $\text{Na}_2\text{O}$  is subsequently added. Explain the observation with the aid of relevant equations.

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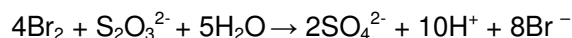
[3]

- (b) Bromine and fluorine can react to form an interhalogen liquid compound  $\text{BrF}_3$ , a common fluorinating agent.

This compound can react with silicon(IV) oxide according to the following equation.



The amount of bromine produced can be determined by reacting it with sodium thiosulfate solution,  $\text{Na}_2\text{S}_2\text{O}_3$  (aq) as shown in the following equation.



- (i) Determine the mass of  $\text{SiF}_4$  produced if  $30.0 \text{ cm}^3$  of  $0.100 \text{ mol dm}^{-3} \text{ Na}_2\text{S}_2\text{O}_3$  solution was required to completely react with the bromine liberated from the reaction between  $\text{BrF}_3$  and  $\text{SiO}_2$ .

[Turn Over]

- 3 (b) (ii)** The bromine produced could also be reacted with cold aqueous sodium hydroxide to form bromide,  $\text{Br}^-$  and bromate(V),  $\text{BrO}_3^-$ , ions.  
State the type of reaction that bromine has undergone and write a balanced equation for the reaction involved.

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[6]

[Total: 9m]

- 4** Natural gas is a vital component of the world's supply of energy. It is one of the cleanest, safest, and most useful of all energy sources.
- (a)** One of the uses of natural gas is the reforming of methane to produce pure hydrogen gas which is widely used in the Haber process to manufacture ammonia.



In an industrial reformer, methane and steam are sealed into a  $2 \text{ dm}^3$  vessel at  $700^\circ\text{C}$ . The number of moles of each gas in the vessel at equilibrium is as follows:

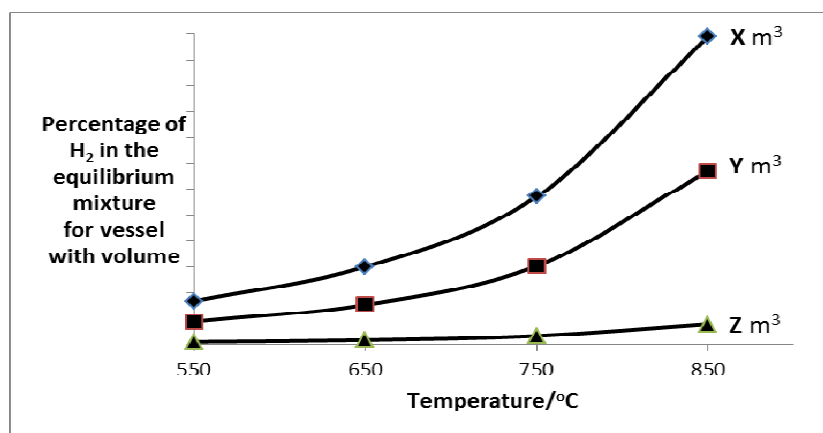
| Gas          | $\text{CH}_4$ | $\text{H}_2\text{O}$ | $\text{H}_2$ | $\text{CO}$ |
|--------------|---------------|----------------------|--------------|-------------|
| No. of moles | 0.24          | 0.18                 | 2.40         | 0.80        |

Calculate the value of the equilibrium constant,  $K_c$ , for the reaction at  $700^\circ\text{C}$ , stating its units.

[2]



- 4 (b) The following graph shows how the percentage of  $H_2$  in the equilibrium mixture varies with temperature and the volumes (**X**, **Y** and **Z**  $m^3$ ) of the reaction vessel in a methane reformer.



Use the data to

- I deduce whether the production of  $H_2$  is an exothermic or endothermic process
- II arrange, in increasing order, the volumes **X**, **Y** and **Z**.

Explain your answer.

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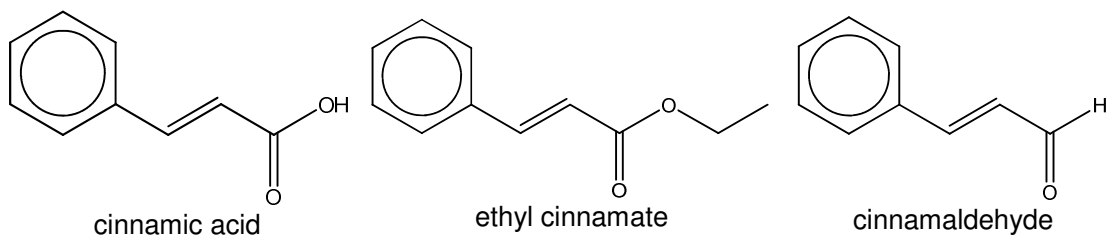
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[4]

[Total: 6m]

[Turn Over]

- 5 Cinnamic acid, ethyl cinnamate and cinnamaldehyde are flavour components in the essential oil of cinnamon.



- (a) Describe, by means of simple chemical tests, how you would distinguish between cinnamaldehyde and ethyl cinnamate.

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[2]

- (b) State the reagents and conditions for converting cinnamic acid to ethyl cinnamate in the laboratory.

.....

[1]

- (c) Cinnamic acid has a honey-like odour while ethyl cinnamate has a fruity odour. State and explain which of the compounds you would expect to be more volatile.

.....  
 .....  
 .....  
 .....

[3]

[Total: 6m]

### Section B (40 marks)

Answer **two out of three** questions in this section on the writing papers provided.

- 6 (a) Explain the following observations with the aid of appropriate diagrams.
- (i) The nitronium ion,  $\text{NO}_2^+$ , is linear, but the nitrite ion,  $\text{NO}_2^-$ , is non-linear.
  - (ii) The solubility in water of hydrazine,  $\text{H}_2\text{N}-\text{NH}_2$ , is considerably higher than that of  $\text{CO}_2$ .
  - (iii) Aluminium oxide exists as a solid whilst aluminium iodide exists as a gas at room temperature and pressure.

[6]

- (b) During the catalytic cracking of long chain and branched alkane molecules, fragments of chained or branched molecules could be formed. An example of these fragments is 2-methylprop-1-ene.

- (i) 2-methylprop-1-ene first undergoes addition reaction with hydrogen gas and nickel catalyst, followed by gaseous chlorine under ultraviolet light to give two different mono-chlorinated products. State the ratio of the products formed and draw the structural formula of each of the two products.
- (ii) Construct a synthetic pathway for the conversion of 2-methylprop-1-ene to ethanoic acid.

[5]

- (c) 2-chloro-2-methylpropan-1-ol, can be formed when aqueous chlorine is added to 2-methylprop-1-ene. 2-chloro-2-methylpropan-1-ol is then reacted with acidified potassium manganate(VII) solution to form compound **HA**.

- (i) Write out the displayed formula of compound **HA**. Hence, write an expression for the acid dissociation constant,  $K_a$ , of **HA**.
- (ii) Explain the difference in the strength of acidity between **HA** and methylpropanoic acid.

[Turn Over]

- 6 (c) (iii) An aqueous solution of an acid  $\text{HA}$  of concentration  $0.010 \text{ mol dm}^{-3}$  has a pH of 3.0. Explain with reasons, whether  $\text{HA}$  is a strong or a weak acid.
- (iv) A solution containing  $\text{HA}$  and its potassium salt  $\text{K}^+\text{A}^-$ , acts as a *buffer* solution. Define the term in *italics*. Hence, write ionic equations to show how this solution reacts with
- (I) Added  $\text{H}^+$  (aq) ions,
- (II) Added  $\text{OH}^-$  (aq) ions.

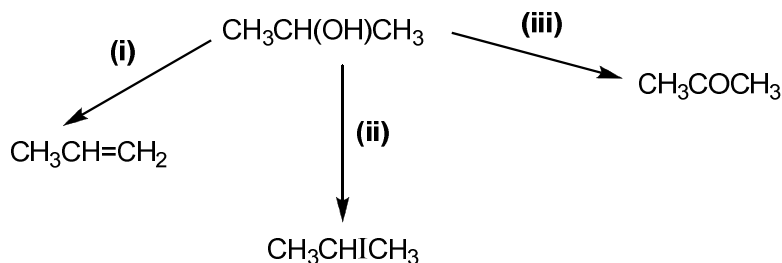
[9]

[Total marks: 20]

- 7 (a)** Drink driving is the act of driving a motor vehicle with blood levels of alcohol in excess of a specified limit. The breathanalyser was invented to measure alcohol content in blood and one of the initial prototype utilises the oxidation process of ethanol using acidified potassium dichromate(VI).
- (i)** State the electronic configuration of chromium in potassium dichromate(VI).
- (ii)** Derive the ion electron half equation for the oxidation of ethanol to ethanoic acid. Hence, using relevant information from the *Data Booklet*, write a balanced equation for the oxidation of ethanol using acidified potassium dichromate(VI).
- (iii)** Using your answer in **(a)(ii)** and given that the breathanalyser contains  $15.0 \text{ cm}^3$  of  $0.017 \text{ mol dm}^{-3}$  acidified potassium dichromate(VI), calculate the minimum volume of vaporised ethanol at room temperature and pressure needed to turn all the potassium dichromate(VI) to green chromium(III) ions.
- (iv)** When potassium dichromate(VI) is reacted with sodium hydroxide, a colour change from orange to yellow was observed due to the formation of potassium chromate(VI),  $\text{K}_2\text{CrO}_4$ .  
Predict and explain the differences in deflection between chromate(VI) and dichromate(VI) ions when passed through an electric field.

**[7]**

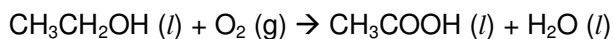
- 7 (b) 2-propanol is thought to be similar to ethanol in its effects on human body, but two to four times more potent. Suggest suitable reagents and conditions for the following reactions of 2-propanol.



- (iv) Hence, use these reactions to describe how you could identify between a sample of 2-propanol and 1-propanol.

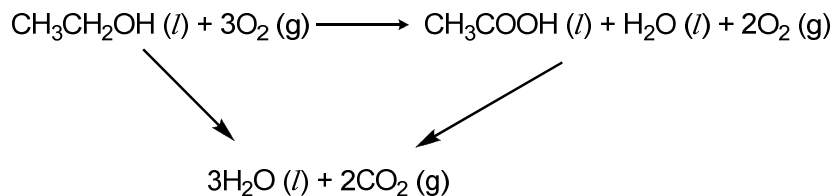
[5]

- (c) Ethanol, upon prolonged exposure to the atmosphere, would eventually oxidise to form ethanoic acid according to the following equation.



Use the following data and energy cycle to calculate the standard enthalpy change for the oxidation of ethanol.

|                                                | $\Delta H^\ominus / \text{kJ mol}^{-1}$ |
|------------------------------------------------|-----------------------------------------|
| Enthalpy change of combustion of ethanol       | -1371                                   |
| Enthalpy change of combustion of ethanoic acid | -876                                    |



[2]

- 7 (d) Geraniol,  $C_9H_{16}O$ , is a primary alcohol used in perfumery. On oxidation of geraniol with hot acidified  $KMnO_4$  (aq), effervescence is observed and two organic products **R**,  $C_3H_6O$ , and **S**,  $C_4H_6O_3$ , are isolated. Both compounds **R** and **S** give an orange precipitate with 2,4-dinitrophenylhydrazine and a yellow precipitate in alkaline aqueous iodine. However, only **S** forms effervescence in aqueous sodium hydrogen carbonate.

Deduce and draw the structural formulae of compounds **R**, **S** and geraniol.

[6]

[Total: 20]

[Turn Over]

- 8 (a) An experiment was set up to determine the mechanism of 2-bromo-3-methylpentane, **FA3**, of concentration  $0.01 \text{ mol dm}^{-3}$  and sodium hydroxide, **FA4**, of concentration  $0.01 \text{ mol dm}^{-3}$ .

The results were tabulated in the following table:

| Experiment number | Vol. of <b>FA3</b> used / $\text{cm}^3$ | Vol. of <b>FA4</b> used / $\text{cm}^3$ | Vol. of water used / $\text{cm}^3$ | Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$ |
|-------------------|-----------------------------------------|-----------------------------------------|------------------------------------|--------------------------------------------|
| 1                 | 10.00                                   | 20.00                                   | 20.00                              | $8.17 \times 10^{-7}$                      |
| 2                 | 25.00                                   | 15.00                                   | 10.00                              | $1.53 \times 10^{-6}$                      |
| 3                 | 30.00                                   | 15.00                                   | 5.00                               | $1.84 \times 10^{-6}$                      |

- (i) Use the data in the table to deduce the order of reaction with respect to 2-bromo-3-methylpentane and sodium hydroxide.
- (ii) Write an overall rate equation for the reaction between sodium hydroxide and 2-bromo-3-methylpentane.
- (iii) By using the collision theory of reaction kinetics, describe briefly the effect on the rate when a catalyst is added into the reaction mixture.

[5]

- (b) 2-bromo-3-methylpentane reacts with hot aqueous sodium hydroxide to give alcohol **G**.

0.04 mol of **G** and 0.02 mol of ethanoic acid were dissolved and made up to  $100 \text{ cm}^3$  and allowed to react in the presence of concentrated sulfuric acid.

At equilibrium,  $10.0 \text{ cm}^3$  of the mixture required  $24.80 \text{ cm}^3$  of  $0.08 \text{ mol dm}^{-3}$  potassium hydroxide for complete neutralisation.

- (i) Write a balanced equation between **G** and ethanoic acid.
- (ii) Calculate the amount of ethanoic acid that reacted with potassium hydroxide. Hence, determine the amount of ethanoic acid at equilibrium in  $100 \text{ cm}^3$  of the mixture.



8 (b) (iii) Hence, calculate the  $K_c$  value for the reaction in (b)(i).

(iv) Given that the reaction is endothermic, explain the effect on the position of the equilibrium with increasing temperature.

[7]

(c) Compound **H**,  $C_{11}H_{12}O_4$ , is a sweet smelling compound that turns moist blue litmus paper red. On boiling **H** with aqueous sodium hydroxide, **I** and **J** are formed.

Product **I** can also be formed by boiling **K** under reflux with silver(I) diammine complex ions (Tollen's reagent). When **K** was reacted with Fehling's solution, no reaction occurred. On acidification of **I**, a white precipitate, **L**,  $C_7H_6O_2$ , was formed.

Product **J**,  $C_4H_7O_3^-$ , does not decolourise purple  $KMnO_4$ . When 0.005 mole of **J** is acidified, it reacts with sodium to form 0.005 mole of hydrogen gas at room temperature and pressure. When **J** is acidified and reacted with potassium carbonate, it gives off an effervescence which forms a white precipitate with limewater.

Deduce the structures of compounds **H**, **I**, **J**, **K** and **L**. Explain the chemistry of the reactions described.

[8]

[Total: 20]

~~~ END OF PAPER ~~~

[Turn Over]